SCOT: Proofs of Formal Results

Companion to SCOT: Queryable Semantic Compression via Complete One-Hop Transformed Rule Set. Numbered statements, algorithms, sections, and equations referenced below use the numbering of the manuscript dated November 12, 2026.

APPENDIX A
PROOFS OF FORMAL RESULTS

We use $\Sigma \models \gamma$ in its standard rule-consequence sense: every model of Σ satisfies every ground instance of γ.

Proof of Lemma 1. Consider one Phase 1 deletion step. Let $G' \subseteq G$ be the current graph and γ the non-recursive rule used in this step. The subgraph deleted in this step is $G_{\mathrm{del}} = G' \cap \tau_{\{\gamma\},G'}$. Every fact in G_{del} has the head predicate of γ, which occurs in no body atom of γ. Consequently, for each $d \in G_{\mathrm{del}}$, any body match in G' witnessing d remains a body match in $G' \setminus G_{\mathrm{del}}$. Thus $G_{\mathrm{del}} \subseteq \Delta_{G' \setminus G_{\mathrm{del}},\Sigma}$ and $G' \subseteq \Delta_{G' \setminus G_{\mathrm{del}},\Sigma}$. Since $G' \setminus G_{\mathrm{del}} \subseteq G'$, monotonicity and idempotence of least-fixpoint closure imply

$$\Delta_{G' \setminus G_{\mathrm{del}},\Sigma} = \Delta_{G',\Sigma}.$$

Applying this argument to every Phase 1 deletion proves that its output G' is a subset of G and has closure $\Delta_{G,\Sigma}$. The argument does not depend on the particular SCC or rule order.

For a Phase 2 round, let G' be its input graph and let $G'_i = \bigcup_{j=1}^{i} G^{(j)}_{\mathrm{blk}}$ be the subgraph formed by its first i blocks, with $G'_0 = \emptyset$. Denote the retained set and its closure after block i by $G^{(i)}_{\mathrm{keep}}$ and $G^{(i)}_{\mathrm{cl}} = \Delta_{G^{(i)}_{\mathrm{keep}},\Sigma}$, with $G^{(0)}_{\mathrm{keep}} = G^{(0)}_{\mathrm{cl}} = \emptyset$. The initial closure is empty because PD rules have nonempty bodies. We claim, by induction on i, that

$$G^{(i)}_{\mathrm{keep}} \subseteq G'_i \subseteq G^{(i)}_{\mathrm{cl}}.$$

It holds for $i = 0$. At block i, the update $G^{(i)}_{\mathrm{keep}} = G^{(i-1)}_{\mathrm{keep}} \cup (G^{(i)}_{\mathrm{blk}} \setminus G^{(i-1)}_{\mathrm{cl}})$ retains only input facts, so $G^{(i)}_{\mathrm{keep}} \subseteq G'_i$. Every fact of G'_{i-1} lies in $G^{(i-1)}_{\mathrm{cl}} \subseteq G^{(i)}_{\mathrm{cl}}$. A fact of $G^{(i)}_{\mathrm{blk}}$ either already lies in $G^{(i-1)}_{\mathrm{cl}}$ or is added to $G^{(i)}_{\mathrm{keep}} \subseteq G^{(i)}_{\mathrm{cl}}$. Hence $G'_i \subseteq G^{(i)}_{\mathrm{cl}}$, proving the induction step. At the last block, the partition gives $G'_{N_b} = G'$, so $G^{(N_b)}_{\mathrm{keep}} \subseteq G' \subseteq \Delta_{G^{(N_b)}_{\mathrm{keep}},\Sigma}$. Monotonicity and idempotence yield $\Delta_{G',\Sigma} = \Delta_{G^{(N_b)}_{\mathrm{keep}},\Sigma}$. Each round therefore preserves its input closure while retaining a subset of its input. Induction over the T rounds, following Phase 1, proves $G^\dagger \subseteq G$ and $\Delta_{G^\dagger,\Sigma} = \Delta_{G,\Sigma}$. All loops terminate because the graphs and rule set are finite and the function-free closure reaches a finite fixpoint. □

Proof of Theorem 1. We first justify the derivation binding returned by Alg. 2. Define $G^{(0)} = G^\dagger$ and $G^{(i)} = G^{(i-1)} \cup \tau_{\Sigma,G^{(i-1)}}$ for $i \geq 1$. At the start of iteration i, the algorithm has $G_{\mathrm{known}} = G^{(i-1)}$ and, for $i > 1$, $G_{\mathrm{new}} = G^{(i-1)} \setminus G^{(i-2)}$; for $i = 1$, $G_{\mathrm{new}} = G^{(0)}$. Every head admitted by the algorithm belongs to $G^{(i)} \setminus G^{(i-1)}$, because all of its body facts belong to $G^{(i-1)}$ and previously known heads are excluded. Conversely, let $d \in G^{(i)} \setminus G^{(i-1)}$ and choose a rule instance deriving it from $G^{(i-1)}$. At least one body occurrence belongs to G_{new}. Otherwise all its body facts would already belong to $G^{(i-2)}$, and d would be in $G^{(i-1)}$. For $i = 1$, every body occurrence belongs to the initial G_{new}. The algorithm examines this position and therefore admits d. It records one witness for each newly admitted head, even if several instances derive the same head. Induction proves that the returned G^* is $\Delta_{G^\dagger,\Sigma}$ and that every $d \in G^* \setminus G^\dagger$ has a recorded witness whose body facts have strictly smaller derivation levels. Finiteness of the active domain makes this process terminate.

Now consider UNFOLDRULE for an omitted fact d. Maintain the following invariant for its current rule $\gamma^{(j)}$ and ground witness σ_{w}:

$$\Sigma \models \gamma^{(j)}, \qquad \sigma_{\mathrm{w}}(H_{\gamma^{(j)}}) = d, \qquad \sigma_{\mathrm{w}}(B_{\gamma^{(j)}}) \subseteq G^*.$$

It holds initially for the freshly renamed input rule recorded for d. If a body occurrence β has witness $e = \sigma_{\mathrm{w}}(\beta) \notin G^\dagger$, the binding $\mathcal{D}[e]$ provides a freshly renamed rule γ' and ground witness σ' with $\sigma'(H_{\gamma'}) = e$. The two clauses have disjoint variables. Their ground witnesses agree on the resolved atom, so their union unifies β and $H_{\gamma'}$. The relevant MGU μ therefore factors this ground unifier on the resolved atoms. Extending the factorization by the original ground assignments on other clause variables gives the ground substitution λ used in the algorithm. Applying μ to the whole rule and replacing just the selected body occurrence by $\mu(B_{\gamma'})$ preserves the witnessed head and assigns the child's recorded body facts to the inserted occurrences. Definite-clause resolution also preserves $\Sigma \models \gamma^{(j+1)}$. Range restriction is preserved: a head variable whose only body occurrence was replaced remains represented in the range-restricted child's body. Thus the invariant continues to hold.

Each expanded occurrence has a derivation level greater than the levels of its inserted supports. Expansion therefore forms a finitely branching tree of depth at most ℓ, the number of productive derivation levels. It terminates with every ground body fact in $G^\dagger$. Keeping occurrences distinct until the end ensures that only the chosen occurrence is expanded. Removing duplicate body atoms on return changes neither the clause's meaning nor its ground witness. Hence the returned path rule γ_d is entailed by Σ and has a ground instance with head d and body contained in $G^\dagger$.

Every initialized identity rule is a tautology and covers its retained facts in one step; pruning retains these identity rules. During the loop in Alg. 3, each omitted fact is either already covered by a current rule or receives such a path rule. Therefore,

before pruning, every fact in G^* is covered by an instance over $G^\dagger$. Deduplication preserves this property. Suppose a remaining rule γ' subsumes a rule γ being removed, witnessed by a substitution θ with $\theta(H_{\gamma'}) = H_\gamma$ and $\theta(B_{\gamma'}) \subseteq B_\gamma$. For any ground witness σ of γ over $G^\dagger$, $\sigma \circ \theta$ witnesses the same head through γ', since

$$(\sigma \circ \theta)(B_{\gamma'}) \subseteq \sigma(B_\gamma) \subseteq G^\dagger.$$

Each pruning step therefore preserves coverage. All surviving rules are entailed by Σ, giving $G^\dagger \cup \tau_{\Sigma^{\mathrm{cot}},G^\dagger} \subseteq \Delta_{G^\dagger,\Sigma}$; coverage gives the reverse inclusion. Thus Σ^{cot} satisfies all conditions of Definition 1. At most one finite path rule is generated per omitted closure fact, in addition to finitely many identities, so Σ^{cot} is finite. None of these operations uses the query. □

Proof of Theorem 2. Lemma 1 gives $G^* = \Delta_{G^\dagger,\Sigma}$, and Theorem 1 gives $G^* = G^\dagger \cup \tau_{\Sigma^{\mathrm{cot}},G^\dagger}$. The initialized identities cover every fact of $G^\dagger$, and the pruning argument in the preceding proof preserves that coverage. Consequently, the stronger atomic equality

$$\tau_{\Sigma^{\mathrm{cot}},G^\dagger} = G^*$$

holds. This equality is useful because atom-wise rewriting uses rule candidates, rather than a separate stored-fact branch.

Fix a query atom α_k with distinct variables $\vec{Z}_k$. Let R_k^* be its match relation over G^*, with columns $\vec{Z}_k$. For any candidate rule, a body match over $G^\dagger$ produces an instance of its head in G^* by soundness. Unifying that head with α_k therefore produces a tuple in R_k^*; the output mapping preserves constants and repeated variables. Conversely, take any tuple in R_k^* and its ground match of α_k. By the atomic equality, a ground instance of some $\gamma \in \Sigma^{\mathrm{cot}}$ has the same head fact and a body match in $G^\dagger$. Rename γ apart from the query. The two head matches give a ground unifier of H_γ and α_k. Relevant-MGU factorization yields a match of the corresponding candidate body, extending the ground substitution to body-only rule variables using the original rule-instance witness. Its output tuple is precisely the selected tuple of R_k^*. Hence the union of candidate branches equals R_k^*.

The query frontier ξ_k contains every variable of α_k that either occurs in another query atom or is an answer variable. All other variables of α_k can be existentially quantified within this group. Therefore its compiled relation is $R_k = \pi_{\xi_k}(R_k^*)$. By ordinary CQ semantics, the answer relation for Q over G^* is

$$\pi_{\vec{X}}\big(R_1 \bowtie \cdots \bowtie R_m\big).$$

This also holds for disconnected atoms, constants, and identified variables. With an empty frontier, R_k is either the unit zero-column tuple or the empty relation, so it contributes exactly the required existence condition. The same convention handles Boolean queries. This proves P00.

For QFSE, fix any ordering $\beta_1, \ldots, \beta_s$ of a candidate body's atom occurrences. The retained variable set $\xi_{k,i}^{(j)}$ in §V-B contains exactly the variables among the first j atoms needed by the output or a later atom. The stage invariant is

$$R_{k,i}^{(j)} = \pi_{\xi_{k,i}^{(j)}}(R_{\beta_1} \bowtie \cdots \bowtie R_{\beta_j}),$$

with $R_{k,i}^{(0)} = \{()\}$. A variable projected out at stage $j-1$ occurs neither in the branch's required output nor in any remaining atom; hence it cannot influence a later join. Joining $R_{k,i}^{(j-1)}$ with R_{β_j} and projecting onto $\xi_{k,i}^{(j)}$ establishes the invariant by induction. If every variable in $\xi_{k,i}^{(j)}$ is already a column of $R_{k,i}^{(j-1)}$, the current atom introduces no required output variable. Under set semantics,

$$\pi_{\xi_{k,i}^{(j)}}(R_{k,i}^{(j-1)} \bowtie R_{\beta_j}) = \pi_{\xi_{k,i}^{(j)}}(R_{k,i}^{(j-1)} \ltimes R_{\beta_j}),$$

so the permitted semi-join implements the same stage. Constant and repeated-variable constraints belong to each atom's match relation. At the final stage, the retained variables and any constant output terms restore exactly the original branch's ξ_k columns. Thus QFSE preserves every group, regardless of the chosen body-atom order.

For UFA, each source group with $\xi_k = \{x\}$ is a unary relation on x. Their natural join is exactly $R_x^{\mathrm{u}} = \bigcap_{k \in \mathcal{I}_x} \pi_x(R_k)$. For a multi-frontier group R_j containing x, replacing it by $R_j \ltimes_x R_x^{\mathrm{u}}$ pushes the same membership constraint into that group. If at least one such group remains, its x column keeps the constraint in the top-level join, so the original singleton-frontier sources can be removed. Applying the filter to every eligible group is idempotent under set semantics. If no multi-frontier group contains x, retaining R_x^{u} when x is an answer variable preserves its possible output values; when x is existential, an $\mathrm{EXISTS}(R_x^{\mathrm{u}})$ guard preserves its required nonemptiness. Every zero-frontier group likewise remains an existence guard. These rules also cover an empty top-level join.

In P11, the R_x^{u} membership test is applied to each candidate using its mapped output value, after that value is bound or immediately if it is a constant. A frontier variable needed by the test is included in the fixed required output of QFSE and cannot be projected away before or after the test. Thus the QFSE and UFA equivalences compose. Alg. 4 emits the unions, joins, projections, semi-joins, and existence guards just proved equivalent, with distinct outputs under the paper's set semantics. Its SQL interpretation uses a faithful null-free encoding of $G^\dagger$ and equality joins. For completed executions, each of P00, P01, P10, and P11 therefore returns exactly the answers to Q over G^*. Selecting one of these equivalent modes with the cost-sensitive policy tree cannot change the answer set. Taking a finite union of CQs with a common answer schema also preserves the equality branch by branch. □

A. Auxiliary Size and Cost Bounds

For completeness, the size bound in (1) follows from the same decreasing-level argument. Each productive iteration of Alg. 2 creates at least one of the $M = |G^* \setminus G^\dagger|$ omitted facts, so $\ell \le M$. A path rule's unfolding tree has branching at most $b = \max(\{1\} \cup \{|B_\gamma| : \gamma \in \Sigma\})$ and depth at most ℓ. It has at

most b^ℓ leaf occurrences, hence at most that many body atoms after deduplication. There are at most M generated path rules, each with one head, and P identities, each with one head and one body atom. Thus

$$L_{\Sigma^{\mathrm{cot}}} \leq 2P + M(1 + b^\ell).$$

Deduplication and subsumption removal can only decrease this count. The estimate establishes finiteness; it is exponential in ℓ when $b > 1$.

The grouped candidate representation stores one group record per query atom, one record per candidate, and each candidate body once. With unary/binary predicates and interned symbols, the output mapping has constant size per candidate. For m atoms, n_k candidates in group k, and total candidate-body length L, this gives $O(m + \sum_k n_k + L)$ words, as stated in §VI. The bound concerns the grouped representation, not emitted SQL or database intermediate results. Compiling four modes, extracting their features, traversing a height-h policy tree, and executing one selected plan gives (9). Counting the retained facts, the head/body atoms of Σ^{cot}, and the policy model gives (10) under the stated bounded-arity and interned-symbol convention.